

More generally, if $\psi_\ell(x)$ and $\psi_m(y)$ are components of fields $\psi_{ab}(x)$ and $\psi_{\tilde{a}\tilde{b}}(y)$ for a particle of spin j , in the irreducible (A, B) and $(\tilde{A}, \tilde{B})$ representations of the homogeneous Lorentz group, then

$$\begin{aligned} P_{ab, \tilde{a}\tilde{b}}(p) = & \sum_{a'b'} \sum_{\tilde{a}'\tilde{b}'} \sum_{\sigma} C_{AB}(j\sigma, a'b') C_{\tilde{A}\tilde{B}}(j\sigma, \tilde{a}'\tilde{b}') \\ & \times \left[\exp(-\theta \hat{p} \cdot \mathbf{J}^{(A)}) \right]_{aa'} \left[\exp(+\theta \hat{p} \cdot \mathbf{J}^{(B)}) \right]_{bb'} \\ & \times \left[\exp(-\theta \hat{p} \cdot \mathbf{J}^{(\tilde{A})}) \right]_{\tilde{a}\tilde{a}'} \left[\exp(+\theta \hat{p} \cdot \mathbf{J}^{(\tilde{B})}) \right]_{\tilde{b}\tilde{b}'}, \end{aligned} \quad (6.2.7)$$

where $\sinh \theta = |\mathbf{p}|/m$, while $a, b, \tilde{a}, \tilde{b}$ run by unit steps from $-A$ to $+A$, $-B$ to $+B$, $-\tilde{A}$ to $+\tilde{A}$, and $-\tilde{B}$ to $+\tilde{B}$, respectively, and likewise for the running indices $a', b', \tilde{a}'$, and $\tilde{b}'$.

Inserting Eqs. (6.2.2) and (6.2.3) in Eq. (6.2.1) yields

$$\begin{aligned} -i\Delta_{\ell m}(x, y) = & \theta(x - y) P_{\ell m} \left(-i \frac{\partial}{\partial x} \right) \Delta_+(x - y) \\ & + \theta(y - x) P_{\ell m} \left(-i \frac{\partial}{\partial x} \right) \Delta_+(y - x), \end{aligned} \quad (6.2.8)$$

where $\Delta_+(x)$ is the function introduced in Chapter 5

$$\Delta_+(x) \equiv (2\pi)^{-3} \int d^3p (2p^0)^{-1} e^{ip \cdot x} \quad (6.2.9)$$

in which p^0 is taken as $+\sqrt{\mathbf{p}^2 + m^2}$.

To go further, we must say a bit about how to extend the definition of the polynomial $P(p)$. Eqs. (6.2.2) and (6.2.3) only define $P(p)$ for four-momenta 'on the mass shell', i.e., with $p^0 = \pm\sqrt{\mathbf{p}^2 + m^2}$. Any polynomial function of such four-momentum can always be taken as linear in p^0 , because any power $(p^0)^{2\nu}$ or $(p^0)^{2\nu+1}$ can be written as $(\mathbf{p}^2 + m^2)^\nu$ or $p^0(\mathbf{p}^2 + m^2)^\nu$, respectively. Thus we can define a polynomial $P^{(L)}(q)$ by the conditions that

$$P^{(L)}(p) = P(p) \quad \left(\text{for } p^0 = \sqrt{\mathbf{p}^2 + m^2} \right), \quad (6.2.10)$$

$$P^{(L)}(q) = P^{(0)}(\mathbf{q}) + q^0 P^{(1)}(\mathbf{q}) \quad (\text{for general } q^\mu),$$

where $P^{(0,1)}$ are polynomials depending only on $\mathbf{q}$. We can now use the relations

$$\frac{\partial}{\partial x^0} \theta(x^0 - y^0) = -\frac{\partial}{\partial x^0} \theta(y^0 - x^0) = \delta(x^0 - y^0) \quad (6.2.11)$$

(recall that $\theta(x)$ has a unit step at x^0 , and is otherwise constant) to move